ION-Sim: A Novel Open-Source Simulation Framework for Intraoperative Neurophysiological Monitoring

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Abstract

The educational pathway for expertise in Intraoperative Neurophysiological Monitoring (IONM) is complex and lengthy. It requires a solid foundation in neurological sciences, neurophysiology, and surgical anatomy. Most importantly, it demands direct familiarity with scenarios across various neurosurgical interventions, including supratentorial and spinal surgery, through at least ten different approaches. Intraoperative neurophysiology must adapt based on specific preoperative tests tailored to the patient. It deploys various methods to collect, analyze, and report signals correlated with the procedure performed.

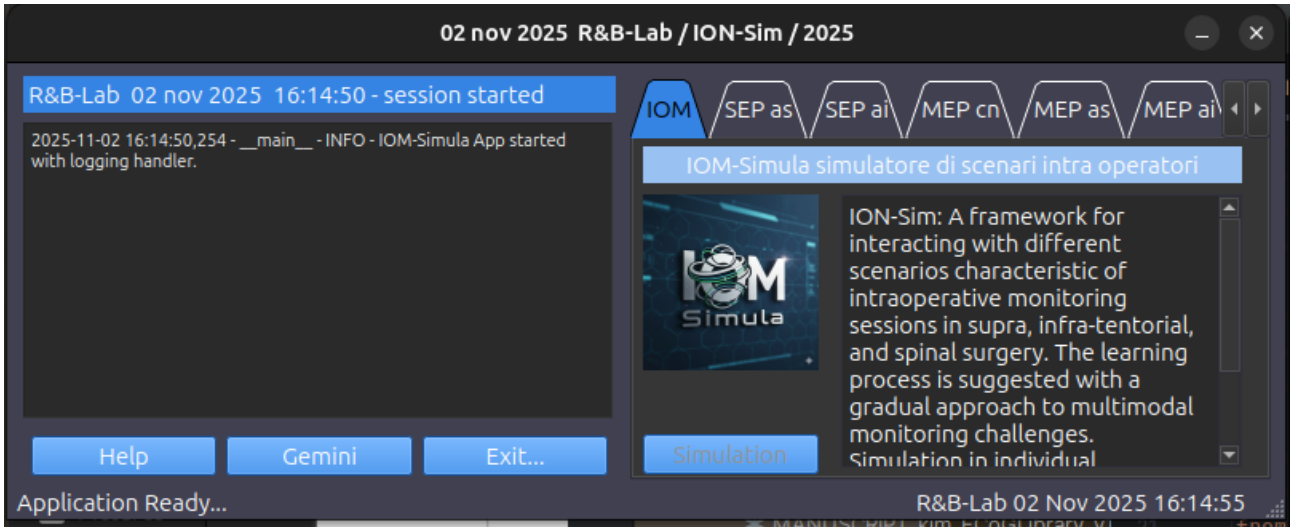
There is a significant shortage of training tools with specific goals for realistic exercise and skill development. The presented project was created in this context and is available upon request for educational purposes.

This project is an open-source framework under the GNU license. It manages a database that covers settings, user data, educational activity results and reports, and tracks objective skill acquisition. The system enables all laboratory testing modalities, adapted to the operating room context. It covers spontaneous EEG and EMG activity, methods for evoked potentials, and techniques for intraoperative stimulus/response tests.

The entire project is available both as a comprehensive Framework and as a scientific Library that can be used, through its specific APIs, to create new configurations and new simulation projects.

Keywords

Medical Education, Simulation, Intraoperative Neurophysiological Monitoring (IONM), Open-Source Software, Computational Neuroscience, Somatosensory Evoked Potentials (SSEP), Motor Evoked Potentials (MEP), Visual Evoked Potential (VEP), Compound Action Potential (CAP), Cortico-Cortical Evoked Potentials (CCEP).



Physiological Simulation Module

The **Physiological Simulation Module** comprises several components, each dedicated to the physiological conditions occurring in specific activation patterns, and includes:

- **CAP (Compound Action Potential) Simulation Module:** Stimulation of the afferent peripheral nerve trunk: sensory fibers recorded at different distances from the stimulation point (wrist, elbow, axilla, Erb's point).
- **CMAP (Compound Motor Action Potential) Simulation Module:** Stimulation of the efferent peripheral nerve trunk: muscle response recorded on different muscle segments at different distances.
- **SEP (Somatosensory Evoked Potential) Simulation Module:** Peripheral nerve stimulation, recording at Erb's point, at the CV7 level, and at the scalp level over the contralateral S1 and M1 sites.
- **VEP (Visual Evoked Potential) Simulation Module:** Flash retinal stimulation, scalp recording over the ipsilateral and contralateral occipital sites.
- **BAEP (Brainstem Acoustic Evoked Potential) Simulation Module:** Click-based acoustic stimulation with contralateral noise, scalp recording - bilateral mastoid.
- **MEPcb (Motor Evoked Potential - Cranio-Bulbar) Simulation Module:** Transcranial scalp stimulation using a pulse train and recording from muscles innervated by cranial nerves (facial, trigeminal, glossopharyngeal, hypoglossal, spinal accessory).
- **MEPas (Motor Evoked Potential - Upper Limbs) Simulation Module:** Muscle activation responses of the upper limbs following transcranial electrical pulse train stimulation: **1st DI** (first dorsal interosseous) muscle, **ECD** (extensor communis digitorum) muscle, **BB** (biceps brachii) muscle.
- **MEPai (Motor Evoked Potential - Lower Limbs) Simulation Module:** Muscle activation responses of the lower limbs following transcranial electrical pulse train stimulation: **FHB** (flexor hallucis brevis) muscle, **TA** (tibialis anterior) muscle, **VL** (vastus lateralis of the quadriceps femoris) muscle.
- **D-Wave (Direct Wave) Simulation Module:** Corticospinal tract activation responses recorded on the spinal cord; stimulation is always transcranial electrical, but recording is performed at two spinal sites, one proximal and one distal to the surgical (or intervention) site.
- **DCS (Direct Cortical Stimulation) Simulation Module:** Responses of various types secondary to direct cortical stimulation (motor responses, language alteration responses, and cognitive function alteration responses).
- **CCEP (Cortico-Cortical Evoked Potentials) Module:** Cortical or subcortical stimulation and recording of the cortical evoked response at a specific distance. Structural connection of tracts and cortical areas.
- **EEG (Electroencephalography) Simulation Module:** Electrical activity at rest and physiological variations for eyes open/closed and motor activity (mu-rhythm).
- **ECoG (Electrocorticography) Module:** Spontaneous corticographical activity with physiological cyclical variations.

- **EMG (Electromyography) Simulation Module:** Spontaneous electromyographic activity with fluctuations and voluntary contraction phenomena during awake patient procedures.

In each of the described modules, the mathematical/physical physiological model for signal generation (peripheral, spinal, cortical, muscle) is implemented, along with the signal propagation model (afferent or efferent pathway), and the nuclear cellular activation model of the spinal cord, thalamus, and the corresponding cortex.

Introduction

Training in clinical neurophysiology is complex in itself and requires the acquisition of practical skills in performing specific tests applied to real patients. This is achieved through defined periods of attendance at accredited training centers affiliated with the Italian Society of Clinical Neurophysiology, which also maintains a dedicated section for Intraoperative Neurophysiology.

Direct learning in the operating room is not always feasible due to organizational issues, and it can be stressful and risky in certain situations. Furthermore, clinical scenarios exhibit considerable variability, and despite the availability of specific protocols for setting up operative session monitoring, there are variants and adaptations necessary for individual cases or patients that can make learning quite uncertain. The learning curve demands a deep understanding of neurophysiology, appropriate pharmacological knowledge—particularly regarding anesthetic procedures—and finally, a strong grasp of cranial and spinal surgical techniques.

Currently, simulation of "intraoperative monitoring" is strictly tied to the use of dedicated equipment, which provides proprietary simulation tools closely linked to the use of that specific apparatus. Naturally, there is a substantial body of literature dedicated to the neurophysiological methodologies used in the operating room, but these instruments generally lack practical interactivity. At present, there are software tools, including open-source ones, for simulating intraoperative scenarios, but these are all applications dedicated to surgical technique, extending even to surgical simulations in VR, yet they do not provide any neurophysiological interaction.

This project aims to bridge some of the gaps outlined by providing educational training through interactive simulations of individual neurophysiological tests and through specific surgical operating room scenarios. In this paper, we present ION-Sim, a novel, open-source, and freely available framework designed to simulate Intra Operative Neurophysiological Monitoring (IONM) signals and clinical scenarios for educational and training purposes. We describe its architecture, functionalities, and demonstrate its utility through representative case studies.

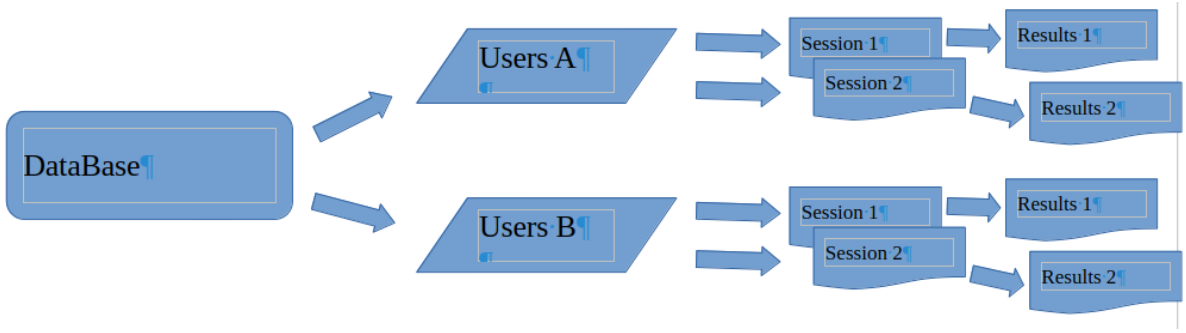
Materials and Methods

The block diagram of the simulation project includes a main block for managing data saved in a database. This module organizes the information that identifies the user, the information that identifies the work and learning sessions, and the information collected from the execution of the proposed simulation modules.

The learning process takes place first through interactive tutorials on the specific features of the simulation module to be used, and subsequently focuses on the utilization of the simulation modules themselves. The learning results, in terms of answers to the proposed questions, are saved in the database and correlated with the session executed.

Finally, a general report will consolidate all the information obtained from the execution of the specific simulation tests.

After entering the user's identification data, the next step is to choose the type of task, or "simulation scenario," that will define the session. These scenarios are detailed in a separate database table, as illustrated in the schema below.



The default simulation scenarios contain the definitions of the test type to be performed and a brief description.

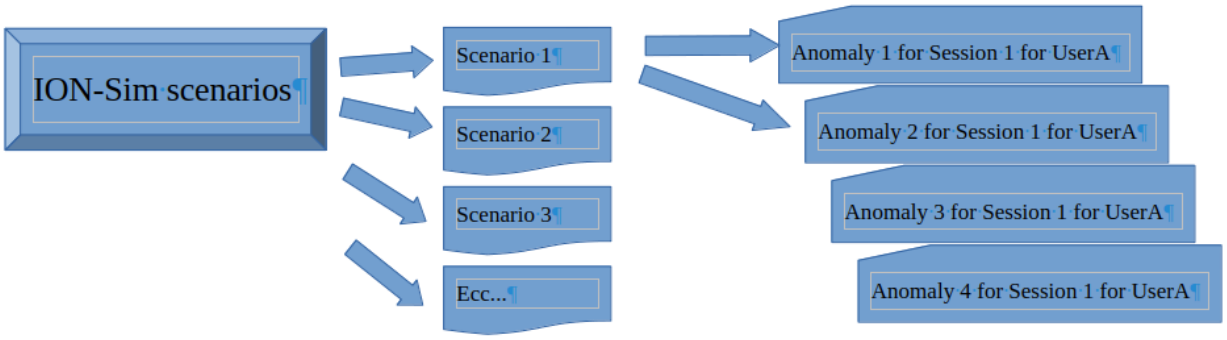
A number of test conditions or 'anomalies' can be associated with each scenario, therefore, every session for every user has a certain number of simulations with associated tests or anomalies. Furthermore, a results table is associated with every session, as previously shown in the schema.

The selectable types of anomalies are logically correlated with the chosen scenario and are provided pre-configured.

The anomalies are automatically chosen by the learning system and presented to the user sequentially, initially all having the same weight during a primary learning phase. Subsequently, the user can choose to repeat or review certain areas of interest by selecting a new session and choosing the type of 'anomaly' they intend to test.

The system also provides for the management of progressive learning with the possibility of activating more complex scenarios and more articulated anomalies. In this case, the user is labeled as 'advanced' and can organize their own learning in a completely personalized manner.

The system also includes the condition of a 'tutor' user, who has the ability to create new test or anomaly conditions based on their own assessments of a specific problem."



System architecture

The simulation framework's architecture begins with a defined database. This framework consists of a main module for database interaction, a series of specialized modules, one for each simulation model and a learning manager to operate the progression of the learning process.
Each simulation module shares a common structure in terms of controls and workflow, but also includes a specific section tailored to its particular scenario. Once a simulation is selected from the database, it initiates automatically. A universal control, present in all scenarios, allows for the insertion of anomalies, also chosen from the database.
Users can operate all described functions independently, or they can be managed by a progressive learning system via learning manager. If the latter is chosen, the user's role is to respond to presented anomalies by selecting the correct answer from a range of alternatives provided through the interface.
Each module is accompanied by a file containing extended help information regarding the simulation model and the neurophysiology concepts involved. This same resource can be accessed using the help button in the main window.